

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Computer Science and Engineering) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Computer Science and Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCOE504B - NUMERICAL METHODS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 Interpolation is the process of estimating values
 - a) Outside the given data range
 - b) At random points
 - c) Using derivatives only
 - d) Within the given data range
- 2 Picard's method is based on
 - a) Differentiation
 - b) Integration
 - c) Finite differences
 - d) Matrix inversion
- 3 The Relaxation method improves convergence by introducing
 - a) Pivot element
 - b) Determinant
 - c) Step size
 - d) Relaxation factor
- 4 Compared to the trapezoidal rule, Simpson's rules generally give
 - a) Same accuracy
 - b) Lower accuracy
 - c) Zero error
 - d) Higher accuracy
- 5 Newton's forward interpolation formula is suitable when the value lies
 - a) Near the end of the table
 - b) Near the beginning of the table
 - c) Outside the table
 - d) At the midpoint
- 6 In the bisection method, the root always lies in an interval where
 - a) $f(a)f(b) < 0$
 - b) $f(a) = f(b)$
 - c) $f'(x) = 0$
 - d) $f(x)$ is linear
- 7 Gauss-Jordan method directly transforms the matrix into
 - a) Upper triangular form
 - b) Diagonal form
 - c) Sparse form
 - d) Lower triangular form
- 8 The Runge-Kutta fourth-order method evaluates the slope
 - a) Once
 - b) Twice
 - c) Three times
 - d) Four times
- 9 The backward difference of a function is defined as
 - a) $f(x) - f(x-h)$
 - b) $f(x+h) - f(x)$
 - c) $f(x+h) - f(x-h)$
 - d) $hf'(x)$
- 10 Newton-Cotes formulas are obtained using
 - a) Polynomial interpolation
 - b) Finite differences
 - c) Differential equations
 - d) Iterative methods
- 11 Euler's method is a _____ numerical method.
 - a) Implicit
 - b) Explicit
 - c) Multi-step
 - d) Predictor-corrector

- 12 Simpson's one-third rule requires the number of subintervals to be
 a) Odd b) Prime c) Even d) Multiple of three

Q2. Solve the following.

- A) Find the root of the equation $x^3 - 4x - 9 = 0$, using the bisection method, correct to 6 three decimal places. 6
- B) Find by Newton's method, the root of the $xe^x - 2 = 0$ equations correct to three decimal places. 6

Q3. Solve the following.

- A) Solve by the Gauss-elimination method. 6
 $10x - 7y + 3z + 5u = 6;$ $-6x + 8y - z - 4u = 5;$
 $3x + y + 4z + 11u = 2;$ $5x - 9y - 2z + 4u = 7.$
- B) Solve the following equations by the Gauss-Jordan method: 6
 $2x - 3y + z = -1;$ $x + 4y + 5z = 25;$ $3x - 4y + z = 2$

Q4. Solve Any Two of the following.

- A) In the following table, the values of y are consecutive terms of a series, of which 12.5 is the 5th term. Find the first and tenth terms of the series. 6
- | | | | | | | | |
|----|-----|-----|------|------|------|------|------|
| x: | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| y: | 2.7 | 6.4 | 12.5 | 21.6 | 34.3 | 51.2 | 72.9 |
- B) Evaluate: 6
 a) $\Delta(x + \cos x)$ b) $\Delta^2 \sin(px + q)$ c) $\Delta(e^{3x} \log 2x)$
- C) Estimate from the following table $f(3.8)$ to three significant figures using Newton 6 backward interpolation formula:
- | | | | | | |
|----|---|-----|-----|-----|-----|
| x: | 0 | 1 | 2 | 3 | 4 |
| y: | 1 | 1.5 | 2.2 | 3.1 | 4.6 |

Q5. Solve Any Two of the following.

- A) Derive the formula for Simpson's 1/3rd rule from Newton Cotes Quadrature 6
 formula and evaluate $\int_0^1 \frac{dx}{x^3 + x + 1}$ using Simpson's 1/3rd rule, choosing a step length of 0.25.
- B) Estimate from the following table $f(3.8)$ to three significant figures using Newton 6 backward interpolation formula:
- | | | | | | |
|----|---|-----|-----|-----|-----|
| x: | 0 | 1 | 2 | 3 | 4 |
| y: | 1 | 1.5 | 2.2 | 3.1 | 4.6 |
- C) Derive the formula for Simpson's Three-Eighth Rule from Newton - Cote's 6 quadrature formula.

Q6. Solve Any Two of the following.

- A) Using the Runge-Kutta method of fourth order, solve $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$ with $y(0) = 1$ at 6
 $x = 0.2$ and $x = 0.4$.
- B) Using Euler's method, solve for y at $x = 0.1$ from $dy/dx = x + y + xy$, $y(0) = 1$, 6
 taking step size $h = 0.025$.

C) Solve the following by Euler's modified method at $x = 1.2$ and 1.4 with $h = 0.2$. 6

$$\frac{dy}{dx} = \log(x+y), y(0) = 2$$

*** End ***